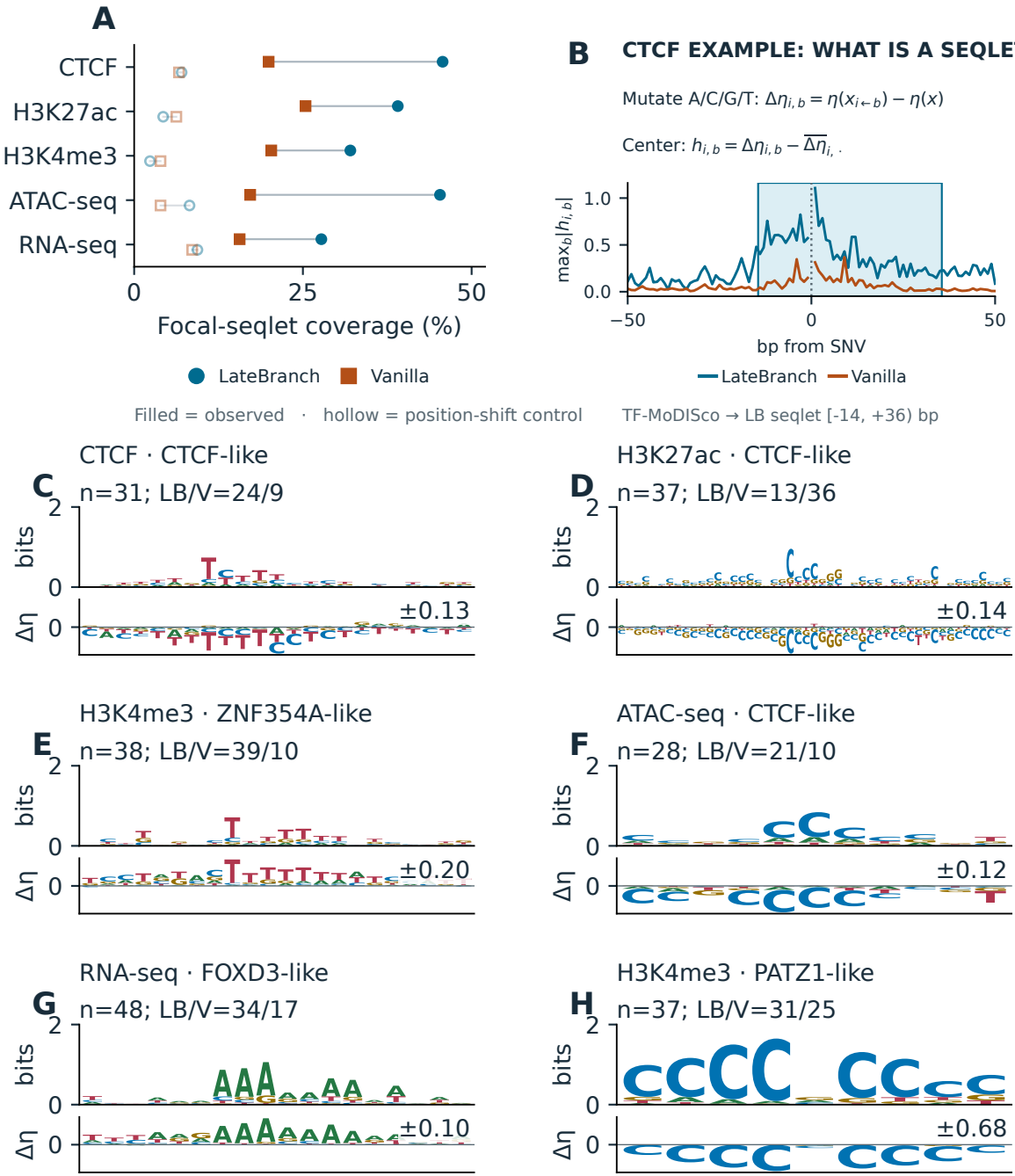

A
CTCF
H3K27ac
H3K4me3
ATAC-seq
RNA-seq
0
25
50
Focal-seqlet coverage (%)
LateBranch
Vanilla
B
CTCF EXAMPLE: WHAT IS A SEQLET?
Mutate A/C/G/T: $\Delta\eta_{i,b} = \eta(x_{i\leftarrow b}) - \eta(x)$
Center: $h_{i,b} = \Delta\eta_{i,b} - \overline{\Delta\eta}_{i,\cdot}$
$\max_b|h_{i,b}|$
1.0
0.5
0.0
−50
0
50
bp from SNV
LateBranch
Vanilla
Filled = observed · hollow = position-shift control
TF-MoDISco → LB seqlet [-14, +36) bp
C
CTCF · CTCF-like
n=31; LB/V=24/9
bits
Δη
±0.13
D
H3K27ac · CTCF-like
n=37; LB/V=13/36
±0.14
E
H3K4me3 · ZNF354A-like
n=38; LB/V=39/10
±0.20
F
ATAC-seq · CTCF-like
n=28; LB/V=21/10
±0.12
G
RNA-seq · FOXD3-like
n=48; LB/V=34/17
±0.10
H
H3K4me3 · PATZ1-like
n=37; LB/V=31/25
±0.68